

1(a)	force per unit mass	B1
1(b)	$GMm/r^2 = m\omega^2$ and $\omega = 2\pi/T$ or $GMm/r^2 = mv^2/r$ and $v = 2\pi r/T$	C1
	$6.67 \times 10^{-11} \times 6.0 \times 10^{24} = r^3 \times [2\pi / (94 \times 60)]^2$	C1
	$r = 6.9 \times 10^6$ m	A1
1(c)(i)	$r^3\omega^2 = \text{constant}$ or $r^3/T^2 = \text{constant}$	C1
	$r^3 / (6.9 \times 10^6)^3 = (150/94)^2$ so $r = 9.4 \times 10^6$ m	A1
	or	
	$GMT^2/4\pi^2 = r^3$ and clear that M is 6.0×10^{24}	(C1)
	$6.67 \times 10^{-11} \times 6.0 \times 10^{24} = r^3 \times [2\pi / (150 \times 60)]^2$ so $r = 9.4 \times 10^6$ m	(A1)
1(c)(ii)	separation increases so (potential energy) increases or movement is against gravitational force so (potential energy) increases	B1
1(c)(iii)	potential energy = $(-)GMm/r$	C1
	$\Delta E_p = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 1200 \times [(6.9 \times 10^6)^{-1} - (9.4 \times 10^6)^{-1}]$	C1
	$= 1.9 \times 10^{10}$ J	A1